

Aalto University
School of Electrical
Engineering

Implementation of a **Virtual Roland Bass Synthesizer TB-303** VST3 Audio Plugin

Mortiz Clus, Giang Vu and María Perolet Lumbierres

Introduction



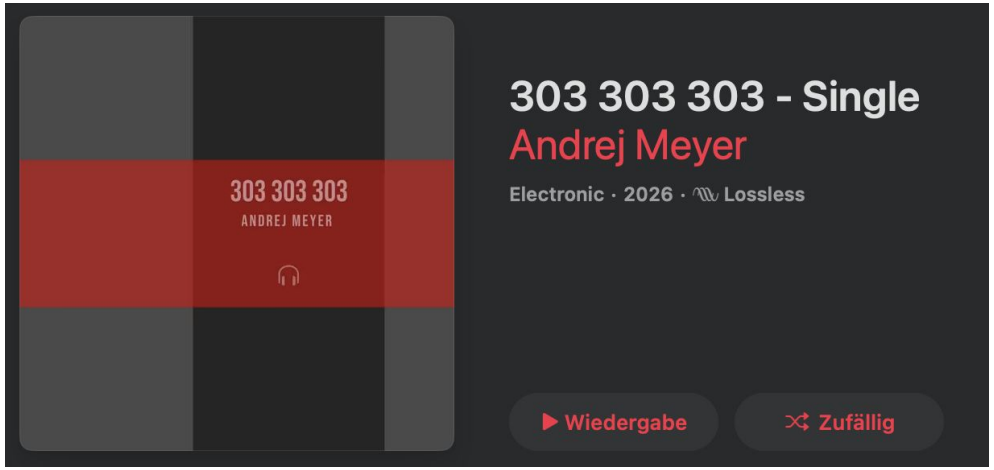


Aalto University

Introduction

- Release in 1981
- Not a commercial success
- Discontinued in 1985
- Re-Discovered by the **Acid-House** Movement
- Evolved into **Acid-Techno**

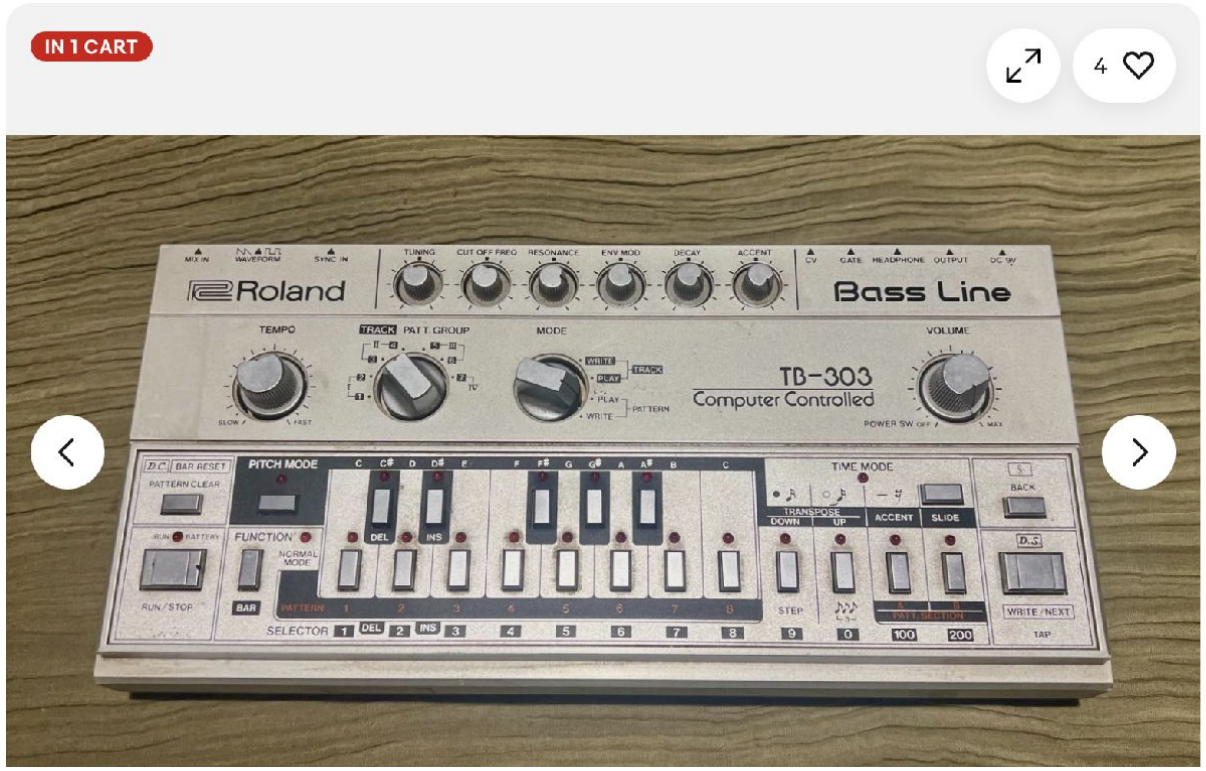




Aalto University

Introduction

- Cult-Status



1981 Roland TB-303 Bassline- show original title

olivar_garage (545)
100% positive · Seller's other items · Message

EUR 3,250.00

Condition: **Used**
"Used but all is OK"

Buy It Now

Add to cart

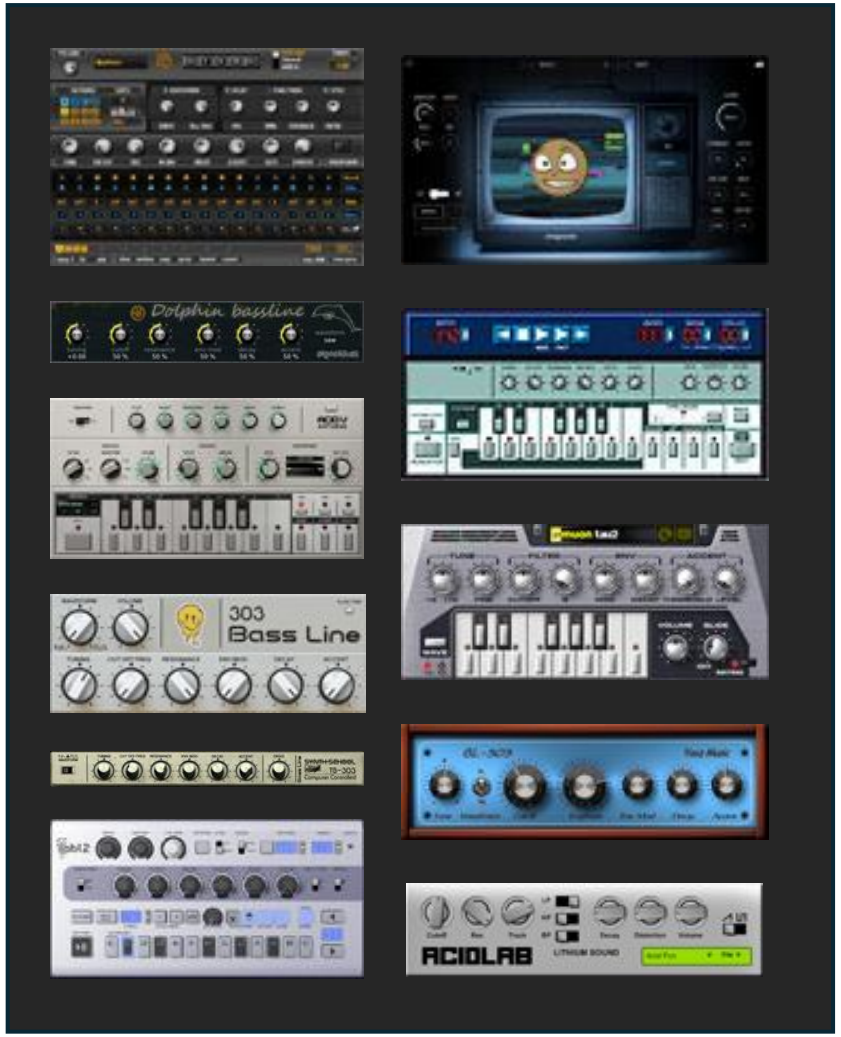
♡ Add to Watchlist

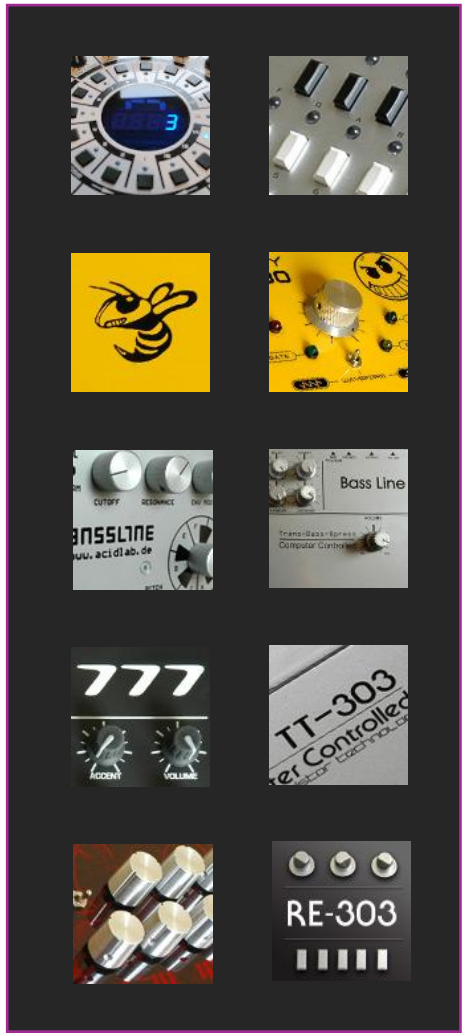
People are checking this out. 4 have added this to their

Aalto University

Introduction

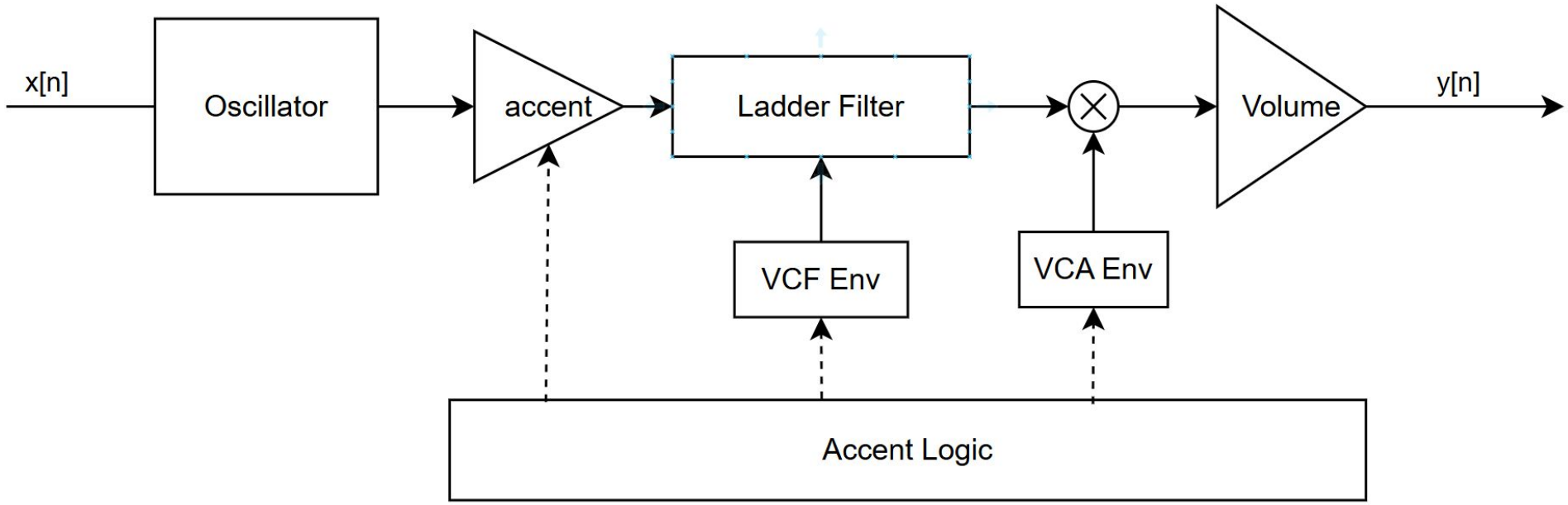
- Cult-Status
- Many HW-Replicas
- Many SW-Replicas





Aalto University

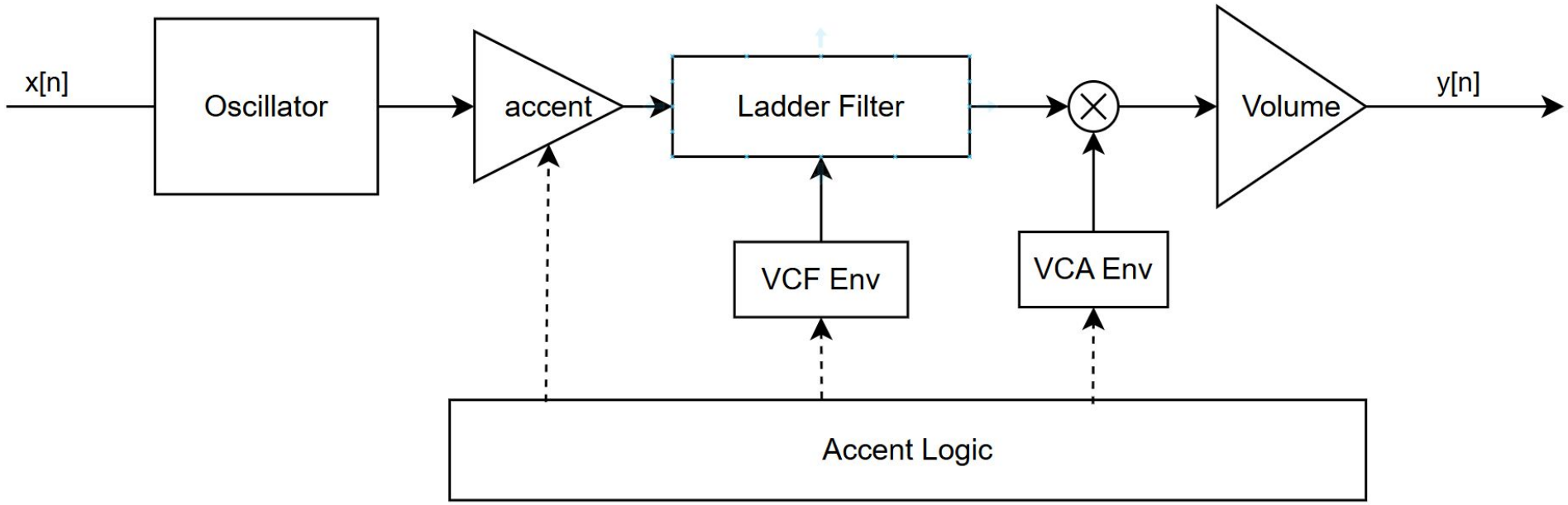
General view



- TB-303 behaves like one tightly-coupled system
- One “accent” control bleeds into several blocks at once

Aalto University

General view



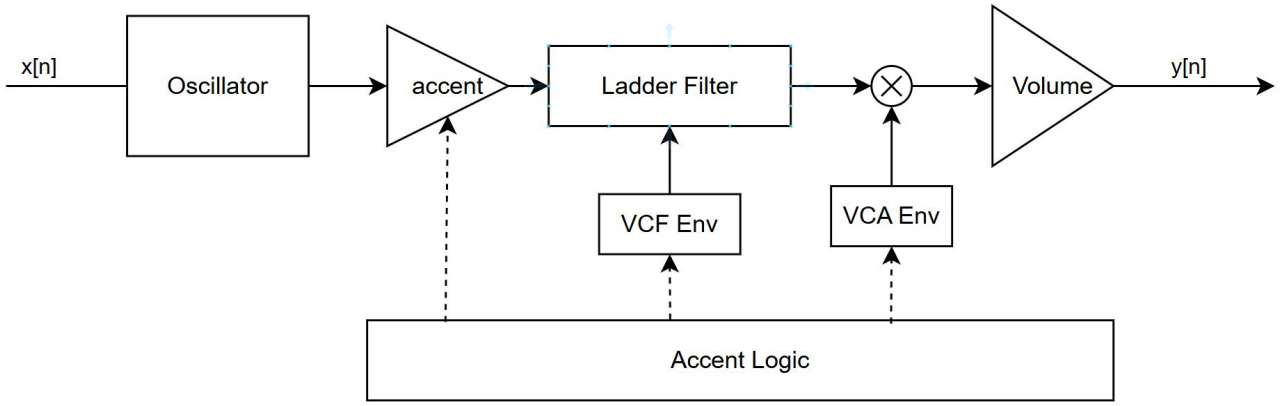
Oscillator → Pre-filter drive →Ladder filter →
VCA (amplifier) →Output

Aalto University

General view

What does the accent do?

- **Pre-filter drive**

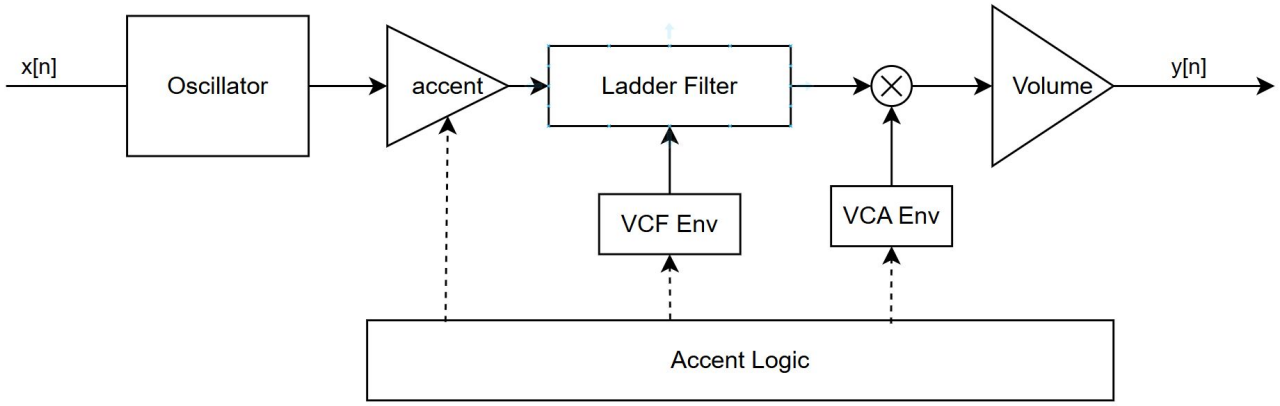


Aalto University

General view

What does the accent do?

- Pre-filter drive
- **Resonance**

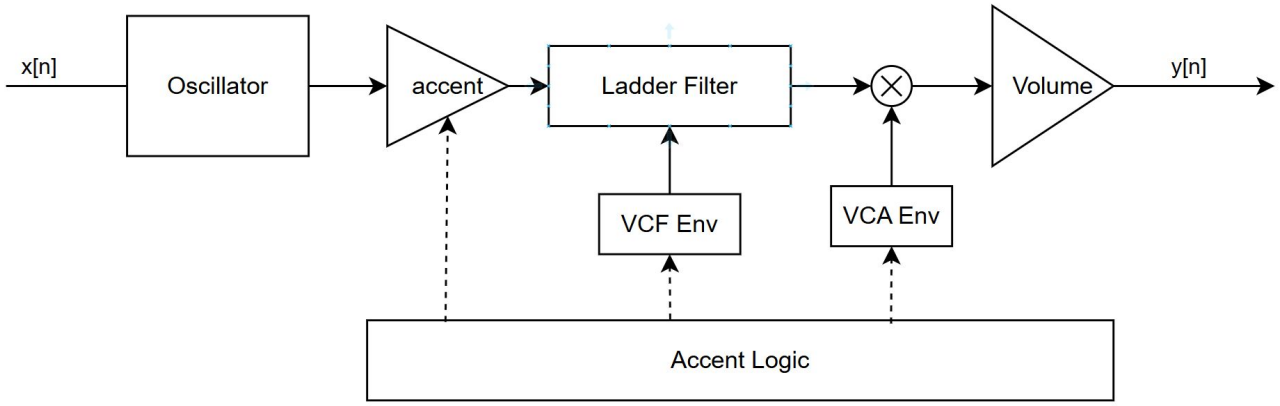


Aalto University

General view

What does the accent do?

- Pre-filter drive
- Resonance
- **Filter envelope depth**

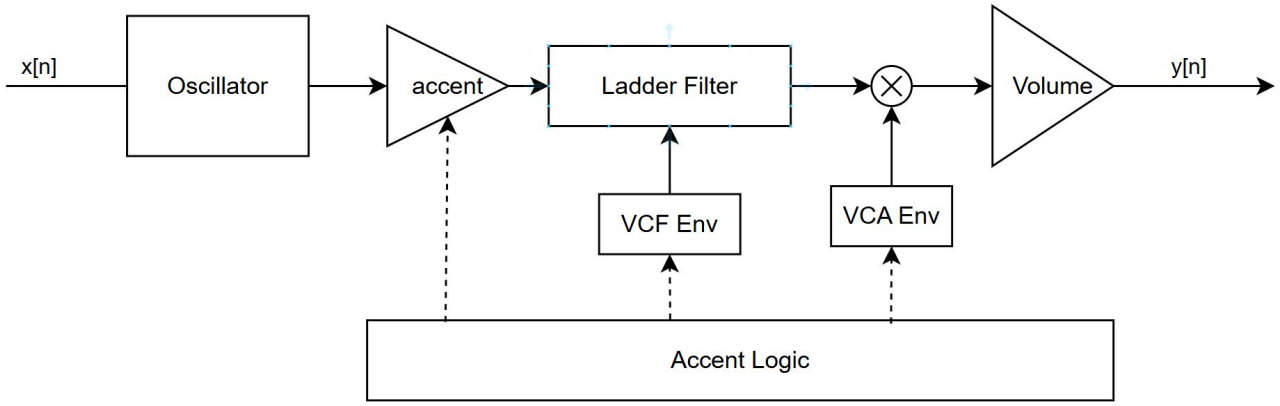


Aalto University

General view

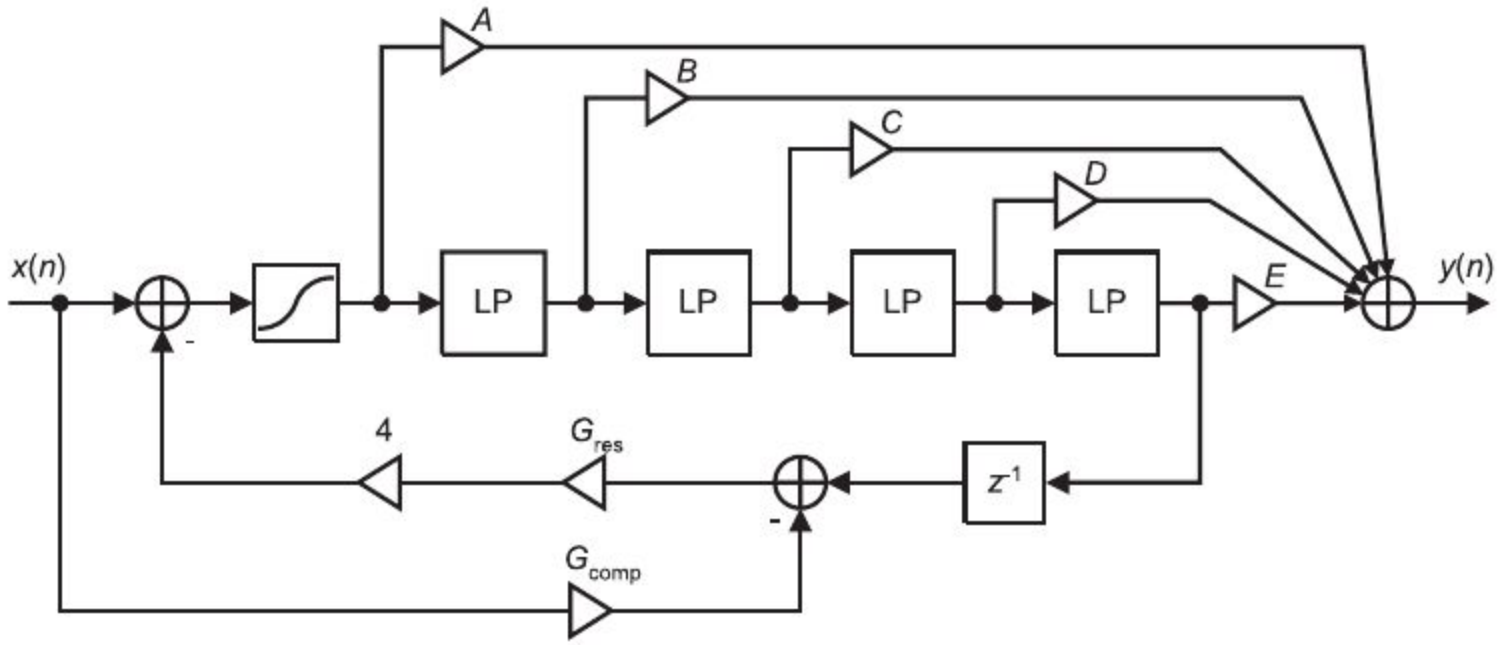
What does the accent do?

- Pre-filter drive
- Resonance
- Filter envelope depth
- **VCA punch**



Aalto University

Ladder Filter



Improved Moog Filter:
Vesa Välimäki and Antti Huovilainen. Oscillator and filter algorithms for virtual analog synthesis. Computer Music Journal, 30:19–31, 06 2006.

Aalto University

Demo

- A / B Testing with Behringer TD-3-MO *(Devilfish Mod Replica)*
- A / B Testing with Bi-Quad Lowpass



Aalto University

Conclusions

- Sound Design is Hard
  - Sounds Cool
  - Not the most accurate
- Achieved Self-Oscillations in Ladder Filter



Aalto University

Future Work

- Parametrize the Envelope Mod
- Add Devilfish-Mod





Aalto University

Thank you!